

МІНІСТЕРСТВО ОСВІТИ І НАУКИ УКРАЇНИ
Національний університет «Полтавська політехніка
імені Юрія Кондратюка»

Кафедра комп'ютерних та інформаційних технологій і систем

Звіт
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з дисципліни «Технології на платформі NET»,
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Додаткове завдання

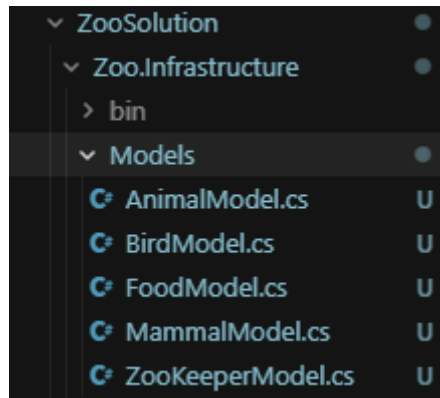


Рис.1 – Створені моделі

```
ZooSolution > Zoo.Infrastructure > ZooContext.cs
1  using Microsoft.EntityFrameworkCore;
2  using Zoo.Infrastructure.Models;
3
4  namespace Zoo.Infrastructure
5  {
6      public class ZooContext : DbContext
7      {
8          public DbSet<AnimalModel> Animals { get; set; }
9          public DbSet<BirdModel> Birds { get; set; }
10         public DbSet<MammalModel> Mammals { get; set; }
11         public DbSet<ZooKeeperModel> ZooKeepers { get; set; }
12         public DbSet<FoodModel> Foods { get; set; }
13
14         public ZooContext(DbContextOptions<ZooContext> options) : base(options){}
15
16         protected override void OnModelCreating(ModelBuilder modelBuilder)
17         {
18             base.OnModelCreating(modelBuilder);
19
20             modelBuilder.Entity<AnimalModel>().ToTable("Animals");
21             modelBuilder.Entity<BirdModel>().ToTable("Birds");
22             modelBuilder.Entity<MammalModel>().ToTable("Mammals");
23
24             modelBuilder.Entity<AnimalModel>()
25                 .HasOne(a => a.Food)
26                 .WithOne(f => f.Animal)
27                 .HasForeignKey<FoodModel>(f => f.AnimalModelId);
28
29             modelBuilder.Entity<ZooKeeperModel>()
30                 .HasMany(z => z.Animals)
31                 .WithOne(a => a.ZooKeeper)
32                 .HasForeignKey(a => a.ZooKeeperModelId);
33         }
34     }
35 }
```

Рис.2 – Контекстна бд

```

ZooSolution > Zoo.Infrastructure > Models > C# AnimalModel.cs
1  namespace Zoo.Infrastructure.Models
2  {
3      public class AnimalModel
4      {
5          public Guid Id { get; set; }
6          public string Name { get; set; } = string.Empty;
7          public int Age { get; set; }
8          public double Weight { get; set; }
9
10         public FoodModel? Food { get; set; }
11
12         public Guid ZooKeeperModelId { get; set; }
13         public ZooKeeperModel? ZooKeeper { get; set; }
14     }
15 }
16
17
18

```

Рис.3 – AnimalModel

```

ZooSolution > Zoo.Infrastructure > Models > C# BirdModel.cs
1  namespace Zoo.Infrastructure.Models
2  {
3      public class BirdModel : AnimalModel
4      {
5          public double Wingspan { get; set; }
6          public bool CanFly { get; set; }
7          public string FeatherColor { get; set; } = string.Empty;
8      }
9  }
10
11
12

```

Рис.4 – BirdModel

```

ZooSolution > Zoo.Infrastructure > Models > FoodModel.cs
1 namespace Zoo.Infrastructure.Models
2 {
3     public class FoodModel
4     {
5         public Guid Id { get; set; }
6         public string Name { get; set; } = string.Empty;
7         public double Quantity { get; set; }
8         public string Unit { get; set; } = string.Empty;
9
10        public Guid AnimalModelId { get; set; }
11        public AnimalModel? Animal { get; set; }
12    }
13 }
14
15
16

```

Рис.5 – FoodModel

```

ZooSolution > Zoo.MVC > {} appsettings.json > ...
1 {
2     "ConnectionStrings": {
3         "DefaultConnection": "Host=localhost;Port=5432;Database=zoo_db;Username=postgres;Password=1234"
4     },
5     "Logging": {
6         "LogLevel": {
7             "Default": "Information",
8             "Microsoft.AspNetCore": "Warning"
9         }
10    },
11    "AllowedHosts": "*"
12 }
13

```

Рис.6 – Файл підключення до бд

```

ZooSolution > Zoo.MVC > Program.cs
1 using Microsoft.EntityFrameworkCore;
2 using Zoo.Infrastructure;
3
4 var builder = WebApplication.CreateBuilder(args);
5
6 builder.Services.AddControllersWithViews();
7 | Ctrl+L to chat, Ctrl+K to generate
8 string? connectionString = builder.Configuration.GetConnectionString("DefaultConnection");
9 builder.Services.AddDbContext<ZooContext>(options =>
10     options.UseNpgsql(connectionString));
11
12 var app = builder.Build();
13
14 if (!app.Environment.IsDevelopment())
15 {
16     app.UseExceptionHandler("/Home/Error");
17     app.UseHsts();
18 }
19
20 app.UseRouting();
21
22 app.UseAuthorization();
23
24 app.MapStaticAssets();
25
26 app.MapControllerRoute(
27     name: "default",
28     pattern: "{controller=Home}/{action=Index}/{id?}"
29     .WithStaticAssets());
30
31
32 app.Run();

```

Рис.7 – Налаштування Program.cs

```

ZooSolution > Zoo.MVC > Controllers > ZooKeepersController.cs

1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Threading.Tasks;
5  using Microsoft.AspNetCore.Mvc;
6  using Microsoft.AspNetCore.Mvc.Rendering;
7  using Microsoft.EntityFrameworkCore;
8  using Zoo.Infrastructure;
9  using Zoo.Infrastructure.Models;
10
11 namespace Zoo.MVC.Controllers
12 {
13     public class ZooKeepersController : Controller
14     {
15         private readonly ZooContext _context;
16
17         public ZooKeepersController(ZooContext context)
18         {
19             _context = context;
20         }
21
22         public async Task<IActionResult> Index()
23         {
24             return View(await _context.ZooKeepers.ToListAsync());
25         }
26
27         public async Task<IActionResult> Details(Guid? id)
28         {
29             if (id == null)
30             {
31                 return NotFound();
32             }
33
34             var zooKeeperModel = await _context.ZooKeepers
35                 .FirstOrDefaultAsync(m => m.Id == id);
36             if (zooKeeperModel == null)

```

Рис.8 – Створення відповідного контролера, частина 1

```

ZooSolution > Zoo.MVC > Controllers > ZooKeepersController.cs
12 {
14 {
61 }
62
63 public async Task<IActionResult> Edit(Guid? id)
64 {
65     if (id == null)
66     {
67         return NotFound();
68     }
69
70     var zooKeeperModel = await _context.ZooKeepers.FindAsync(id);
71     if (zooKeeperModel == null)
72     {
73         return NotFound();
74     }
75     return View(zooKeeperModel);
76 }
77
78 [HttpPost]
79 [ValidateAntiForgeryToken]
80 public async Task<IActionResult> Edit(Guid id, [Bind("Id,FullName,ExperienceYears,Shift")] ZooKeeperModel zooKeeperModel)
81 {
82     if (id != zooKeeperModel.Id)
83     {
84         return NotFound();
85     }
86
87     if (ModelState.IsValid)
88     {
89         try
90         {
91             _context.Update(zooKeeperModel);
92             await _context.SaveChangesAsync();
93         }
94         catch (DbUpdateConcurrencyException)

```

Рис.9 – Створення відповідного контролера, частина 2

```

ZooSolution > Zoo.MVC > Controllers > ZooKeepersController.cs
12 {
14 {
108 }
109
110 public async Task<IActionResult> Delete(Guid? id)
111 {
112     if (id == null)
113     {
114         return NotFound();
115     }
116
117     var zooKeeperModel = await _context.ZooKeepers
118         .FirstOrDefaultAsync(m => m.Id == id);
119     if (zooKeeperModel == null)
120     {
121         return NotFound();
122     }
123
124     return View(zooKeeperModel);
125 }
126 Ctrl+L to chat. Ctrl+K to generate
127 [HttpPost, ActionName("Delete")]
128 [ValidateAntiForgeryToken]
129 public async Task<IActionResult> DeleteConfirmed(Guid id)
130 {
131     var zooKeeperModel = await _context.ZooKeepers.FindAsync(id);
132     if (zooKeeperModel != null)
133     {
134         _context.ZooKeepers.Remove(zooKeeperModel);
135     }
136
137     await _context.SaveChangesAsync();
138     return RedirectToAction(nameof(Index));
139 }
140
141 private bool ZooKeeperModelExists(Guid id)

```

Рис.10 – Створення відповідного контролера, частина 3

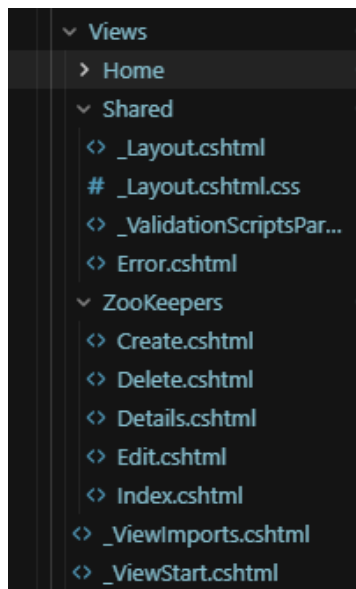


Рис.11 – Вигляд створених cshtml файлів

```

ZooSolution > Zoo.MVC > Views > ZooKeepers > <> Create.cshtml
1  @model Zoo.Infrastructure.Models.ZooKeeperModel
2
3  @{
4      ViewData["Title"] = "Створити";
5  }
6
7  <h1>Створити</h1>
8
9  <h4>Доглядач</h4>
10 <hr />
11 <div class="row">
12     <div class="col-md-4">
13         <form asp-action="Create">
14             <div asp-validation-summary="ModelOnly" class="text-danger"></div>
15             <div class="form-group">
16                 <label asp-for="FullName" class="control-label"></label>
17                 <input asp-for="FullName" class="form-control" />
18                 <span asp-validation-for="FullName" class="text-danger"></span>
19             </div>
20             <div class="form-group">
21                 <label asp-for="ExperienceYears" class="control-label"></label>
22                 <input asp-for="ExperienceYears" class="form-control" />
23                 <span asp-validation-for="ExperienceYears" class="text-danger"></span>
24             </div>
25             <div class="form-group">
26                 <label asp-for="Shift" class="control-label"></label>
27                 <input asp-for="Shift" class="form-control" />
28                 <span asp-validation-for="Shift" class="text-danger"></span>
29             </div>
30             <div class="form-group">
31                 <input type="submit" value="Створити" class="btn btn-primary" />
32             </div>
33         </form>
34     </div>
35 </div>
36
37 <div>
38     <a asp-action="Index">Повернутися до списку</a>
39 </div>

```

Рис.12 – Вигляд файлу Create.cshtml

```

ZooSolution > Zoo.MVC > Views > ZooKeepers > <> Details.cshtml

1  @model Zoo.Infrastructure.Models.ZooKeeperModel
2
3  @{
4      ViewData["Title"] = "Детали";
5  }
6
7  <h1>Детали</h1>
8
9  <div>
10     <h4>Доглядач</h4>
11     <hr />
12     <dl class="row">
13         <dt class = "col-sm-2">
14             @Html.DisplayNameFor(model => model.FullName)
15         </dt>
16         <dd class = "col-sm-10">
17             @Html.DisplayFor(model => model.FullName)
18         </dd>
19         <dt class = "col-sm-2">
20             @Html.DisplayNameFor(model => model.ExperienceYears)
21         </dt>
22         <dd class = "col-sm-10">
23             @Html.DisplayFor(model => model.ExperienceYears)
24         </dd>
25         <dt class = "col-sm-2">
26             @Html.DisplayNameFor(model => model.Shift)
27         </dt>
28         <dd class = "col-sm-10">
29             @Html.DisplayFor(model => model.Shift)
30         </dd>
31     </dl>
32 </div>
33 <div>
34     <a asp-action="Edit" asp-route-id="@Model.Id">Редагувати</a> |
35     <a asp-action="Index">Повернутися до списку</a>
36 </div>
37
38
39

```

Рис.13 – Вигляд файлу Details.cshtml

Welcome

Learn about [building Web apps with ASP.NET Core](#).

Рис.14 – Вигляд сторінки, частина 1

StackQueryCookiesHeadersRouting

Endpoint

Name	Value
Display Name	Zoo.MVC.Controllers.ZooKeepersController.Index (Zoo.MVC)
Route Pattern	{controller=Home}/{action=Index}/{id?}
Route Order	1

Endpoint Metadata

Type	Detail
NullableContextAttribute	System.Runtime.CompilerServices.NullableContextAttribute
NullableAttribute	System.Runtime.CompilerServices.NullableAttribute
ControllerAttribute	Microsoft.AspNetCore.Mvc.ControllerAttribute
AsyncStateMachineAttribute	System.Runtime.CompilerServices.AsyncStateMachineAttribute
DebuggerStepThroughAttribute	System.Diagnostics.DebuggerStepThroughAttribute
ControllerActionDescriptor	Microsoft.AspNetCore.Mvc.Controllers.ControllerActionDescriptor
DataTokensMetadata	DataTokens:
RouteNameMetadata	RouteName: default
ControllerActionFilter	Microsoft.AspNetCore.Mvc.Filters.ControllerActionFilter
SaveTempDataAttribute	Microsoft.AspNetCore.Mvc.ViewFeatures.SaveTempDataAttribute
UnsupportedContentTypeFilter	Microsoft.AspNetCore.Mvc.ModelBinding.UnsupportedContentTypeFilter
SuppressLinkGenerationMetadata	SuppressLinkGeneration: True
RouteDiagnosticsMetadata	Route: {controller=Home}/{action=Index}/{id?}

Route Values

Variable	Value
action	Index
controller	ZooKeepers

Рис.15 – Вигляд сторінки, частина 2

StackQueryCookies**Headers**Routing

Variable	Value
Accept	text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding	gzip, deflate, br, zstd
Accept-Language	uk,en;q=0.9,en-GB;q=0.8,en-US;q=0.7
Cache-Control	max-age=0
Connection	keep-alive
Cookie	Webstorm-de55ace5=c880be19-9424-4a4b-b450-7935df7ef476
Host	localhost:5216
sec-ch-ua	"Microsoft Edge";v="143", "Chromium";v="143", "Not A(Brand";v="24"
sec-ch-ua-mobile	?0
sec-ch-ua-platform	"Windows"
Sec-Fetch-Dest	document
Sec-Fetch-Mode	navigate
Sec-Fetch-Site	none
Sec-Fetch-User	?1
Upgrade-Insecure-Requests	1
User-Agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/143.0.0.0 Safari/537.36 Edg/143.0.0.0

Рис.15 – Вигляд сторінки, частина 3